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Protection device structures with a discontinuous isolation well

Abstract

Device structures with an isolation well and methods of forming a device structure with an isolation well. The structure comprises a first well of a first conductivity type in a semiconductor substrate, and a second well of a second conductivity type in the semiconductor substrate. The second conductivity type is opposite to the first conductivity type. The first well includes a plurality of segments, and the second well is positioned in a vertical direction between the segments of the first well and a top surface of the semiconductor substrate.

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Background/Summary

BACKGROUND

(1) The disclosure relates generally to semiconductor devices and integrated circuit fabrication and, more specifically, to device structures with an isolation well and methods of forming a device structure with an isolation well.

(2) An integrated circuit may be exposed to random electrostatic discharge (ESD) events that can direct potentially large and damaging ESD currents to the sensitive devices of the integrated circuit. An ESD event refers to an unpredictable electrical discharge of a positive or negative current over a short duration and during which a large amount of current is directed to the integrated circuit. An ESD event may occur during post-manufacture chip handling or after chip installation on a circuit

board or other carrier. The high current may originate from a variety of sources, such as the human body, a machine component, or a chip carrier.

(3) Precautions may be taken to protect the integrated circuit from an ESD event. One such precaution is to incorporate an on-chip protection circuit that is designed to avert damage to the sensitive devices of the integrated circuit during an ESD event. If an ESD event occurs, a protection device of the protection circuit is triggered to enter a low-impedance state that conducts the ESD current to ground and shunts the ESD current away from the integrated circuit. The protection device remains clamped in its low-impedance state until the ESD current is drained and the ESD voltage is discharged to an acceptable level.

(4) A common type of protection device commonly deployed in an ESD protection circuit is a silicon-controlled rectifier (SCR). In its quiescent state, the SCR restricts current conduction to leakage current. However, a voltage pulse exceeding an engineered threshold, known as the trigger voltage, initiates the conduction of a forward current between the anode and cathode of the SCR. Even after the trigger voltage is removed, the SCR remains clamped to conduct the forward current so long as the forward current remains above an engineered holding current. When the forward current from the ESD event drops below the holding current, the SCR returns to its quiescent state.

(5) The protection device may be formed with a triple-well layer that is used to electrically isolate the device well from the substrate. In a triple-well technology, the addition of a high-energy implant to the process flow allows for the creation of a separate continuous tub of doped semiconductor material that is junction isolated from the substrate.

(6) Improved device structures with an isolation well and methods of forming a device structure with an isolation well are needed.

SUMMARY

(7) In an embodiment, a structure comprises a semiconductor substrate having a top surface, a first well of a first conductivity type in the semiconductor substrate, and a second well of a second conductivity type in the semiconductor substrate. The second conductivity type is opposite to the first conductivity type. The first well includes a plurality of segments, and the second well is positioned in a vertical direction between the segments of the first well and the top surface of the semiconductor substrate.

(8) In an embodiment, a method comprises forming a first well of a first conductivity type in a semiconductor substrate, and forming a second well of a second conductivity type in the semiconductor substrate. The second conductivity type is opposite to the first conductivity type. The first well includes a plurality of segments, and the second well is positioned in a vertical direction between the segments of the first well and a top surface of the semiconductor substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various embodiments of the invention and, together with a general description of the invention given above and the detailed description of the embodiments given below, serve to explain the embodiments of the invention. In the drawings, like reference numerals are used to indicate like features in the various views.

(2) FIG. 1 is a top view of a structure in accordance with embodiments of the invention.

(3) FIG. 2 is a cross-sectional view taken generally along line 2-2 in FIG. 1.

(4) FIG. 3 is a cross-sectional view of the structure at a fabrication stage subsequent to FIG. 2.

(5) FIG. 4 is a top view of a structure in accordance with alternative embodiments of the invention.

(6) FIG. 4A is a cross-sectional view taken generally along line 4A-4A in FIG. 4.

(7) FIG. 5 is a top view of a structure in accordance with alternative embodiments of the invention.

(8) FIG. 6 is a cross-sectional view of a structure in accordance with alternative embodiments of the invention.

(9) FIG. 7 is a cross-sectional view of a structure in accordance with alternative embodiments of the invention.

(10) FIG. 8 is a cross-sectional view of a structure in accordance with alternative embodiments of the invention.

DETAILED DESCRIPTION

(11) With reference to FIG. 1 and in accordance with embodiments of the invention, a structure **10** for a silicon-controlled rectifier includes a semiconductor substrate **12** that may be comprised of a semiconductor material, such as single-crystal silicon. In an embodiment, the semiconductor substrate **12** may be doped to have p-type conductivity. The semiconductor substrate **12** has a top surface **11** and a bottom surface opposite to the top surface **11** in a vertical direction.

(12) The structure **10** includes a segmented deep well **14** that is formed in the semiconductor substrate **12** over a given depth range beneath the top surface **11**. The deep well **14** includes segments **16** that are doped to have an opposite conductivity type from the semiconductor substrate **12**. The deep well **14** may be formed by introducing a dopant by, for example, ion implantation of the semiconductor substrate **12**. A patterned implantation mask may be formed to define selected areas on the top surface **11** of the semiconductor substrate **12** that are exposed for implantation of ions to form the segments **16**. The implantation mask may include a layer of an organic photoresist that is applied and patterned to form openings exposing the selected areas on the top surface **11** of the semiconductor substrate **12** and determining, at least in part, the location and horizontal dimensions of the segments **16**. The implantation mask has a thickness and stopping power sufficient to block the implantation of ions in masked areas. Implantation conditions (e.g., ion species, dose, kinetic energy) may be selected to tune the electrical and physical characteristics of the deep well **14**. In an embodiment, the deep well **14** may be doped with a concentration of an n-type dopant (e.g., arsenic or phosphorus) to provide n-type conductivity. In an embodiment, the deep well **14** may contain an n-type dopant concentration in a range of about $5 \times 10^{15} \text{ cm}^{-3}$ to about $5 \times 10^{16} \text{ cm}^{-3}$.

(13) The segments **16** of the deep well **14** are discontinuous (i.e., non-continuous) in that only discrete regions of the semiconductor substrate **12** are doped. The deep well **14** is marked by breaks or interruptions defined by distinct and unconnected doped regions having the form of the segments **16**. The segments **16** are buried beneath the top surface **11** of the semiconductor substrate **12** in that an unimplanted thickness of the semiconductor substrate **12** separates the segments **16** from the top surface **11**.

(14) In the representative embodiment, each segment **16** may be a strip that is aligned along a longitudinal axis **18** and that has a width **W** in a direction transverse to the longitudinal axis **18**. In an embodiment, the longitudinal axes **18** of the different segments **16** may have a parallel alignment. The segments **16** have opposite side edges **15**, and the side edges **15** of adjacent pairs of the segments **16** are separated by a gap **20**. The segments **16**, which have a side-by-side arrangement, are islands of doped semiconductor material that are not connected to each other. In that regard, strips of the oppositely-doped semiconductor substrate **12** are disposed in the gaps **20** between adjacent segments **16** such that the strips provided by the segments **16** and the strips provided by the semiconductor substrate **12** alternate with position in a lateral direction transverse to the longitudinal axes **18**. In an embodiment, the width dimension of the gaps **20** may be uniform. In an embodiment, the width dimension of the gaps **20** may be non-uniform. In an embodiment, the segments **16** may have a uniform pitch and a uniform line width. Alternatively, the pitch and/or line width of the segments **16** may be non-uniform. In an embodiment, the segments **16** may have equal or substantially equal lengths.

(15) With reference to FIG. 3 and at a fabrication stage subsequent to FIG. 2, shallow trench isolation regions **22** are formed in the semiconductor substrate **12**. The shallow trench isolation

regions **22** may be formed by patterning shallow trenches in the semiconductor substrate **12** with lithography and etching processes, depositing a dielectric material, such as silicon dioxide, to fill the shallow trenches, and planarizing and/or recessing the deposited dielectric material.

(16) Wells **24, 26** are formed in the semiconductor substrate **12**. The wells **24, 26** are doped to have an opposite conductivity type from the deep well **14**. The wells **24, 26** may be formed by introducing a dopant of a given conductivity type by, for example, ion implantation into the semiconductor substrate **12**. A patterned implantation mask may be formed to define selected areas on the top surface **11** of the semiconductor substrate **12** that are exposed for implantation of ions to form the wells **24, 26**. The implantation mask may include a layer of an organic photoresist that is applied and patterned to form openings exposing the selected areas on the top surface **11** of the semiconductor substrate **12** and determining, at least in part, the location and horizontal dimensions of the wells **24, 26**. The implantation mask has a thickness and stopping power sufficient to block the implantation of ions in masked areas. The implantation conditions (e.g., ion species, dose, kinetic energy) may be selected to tune the electrical and physical characteristics of the wells **24, 26**. In an embodiment, the wells **24, 26** may be doped with a concentration of a p-type dopant (e.g., boron) to provide p-type conductivity.

(17) Wells **28, 30, 32** are formed in the semiconductor substrate **12**. The wells **28, 30, 32** may be formed by introducing a dopant of a given conductivity type by, for example, ion implantation into the semiconductor substrate **12**. A patterned implantation mask may be formed to define selected areas on the top surface **11** of the semiconductor substrate **12** that are exposed for implantation of ions to form the wells **28, 30, 32**. The implantation mask may include a layer of an organic photoresist that is applied and patterned to form openings exposing the selected areas on the top surface **11** of the semiconductor substrate **12** and determining, at least in part, the location and horizontal dimensions of the wells **28, 30, 32**. The implantation mask has a thickness and stopping power sufficient to block the implantation of ions in masked areas. The implantation conditions (e.g., ion species, dose, kinetic energy) may be selected to tune the electrical and physical characteristics of the wells **28, 30, 32**. In an embodiment, the wells **28, 30, 32** may be doped with a concentration of an n-type dopant (e.g., arsenic or phosphorus) to provide n-type conductivity.

(18) The well **26** is positioned in a vertical direction between the deep well **14** and the top surface **11** of the semiconductor substrate **12**. The segments **16** of the deep well **14** function to electrically isolate the well **26** from the semiconductor substrate **12**. The well **26**, which has an opposite conductivity type from the segments **16**, may overlap with some of the segments **16**. The wells **30, 32**, which have the same conductivity type as the segments **16**, may overlap with and adjoin segments **16** at the periphery of the deep well **14**.

(19) Doped regions **34, 36** are formed adjacent to the top surface **11** of the semiconductor substrate **12**. The doped region **34** is positioned in the well **24** and the doped region **36** is positioned in the well **26**. The doped regions **34, 36** are doped to have the same conductivity type as the wells **24, 26** but at a higher dopant concentration. The doped regions **34, 36** are accessible at the top surface **11** of the semiconductor substrate **12**. In an embodiment, the doped regions **34, 36** may be doped (e.g., heavily doped) with a concentration of a p-type dopant (e.g., boron) to provide p-type conductivity. The doped regions **34, 36** may be concurrently formed by selectively implanting ions, such as ions including the p-type dopant, under given implantation conditions with an implantation mask having openings defining the intended locations for the doped regions **34, 36** in the semiconductor substrate **12**.

(20) Doped regions **37, 38, 39, 40** are formed adjacent to the top surface **11** of the semiconductor substrate **12**. The doped region **38** is positioned in the well **24**, and the doped region **40** is positioned in the well **26**. The doped regions **38, 40** are doped to have an opposite conductivity type from the wells **24, 26**. The doped region **37** is positioned in the well **28**, and the doped region **39** is positioned in the well **30**. The doped regions **37, 39** are doped to have the same conductivity type as the wells **28, 30** and may be respectively used to contact the wells **28, 30**. In an

embodiment, the doped regions **37, 38, 39, 40** may be doped (e.g., heavily doped) with a concentration of an n-type dopant (e.g., arsenic or phosphorus) to provide n-type conductivity. The doped regions **37, 38, 39, 40** may be concurrently formed by selectively implanting ions, such as ions including the n-type dopant, under given implantation conditions with an implantation mask having openings defining the intended locations for the doped regions **37, 38, 39, 40** in the semiconductor substrate **12**.

(21) Middle-of-line processing and back-end-of-line processing follow to form an interconnect structure with electrical connections coupled to the structure **10**. The doped regions **34, 38** may be interconnected by an electrical connection **46** in the interconnect structure to define a terminal **42** of the structure **10**, and the doped regions **36, 40** may be interconnected by an electrical connection **46** in the interconnect structure to define another terminal **44** of the structure **10**. The doped region **37, 39** may be connected by electrical connections **45** in the interconnect structure to either ground or a bias voltage.

(22) The structure **10** may embody a bidirectional device structure for a silicon-controlled rectifier that may be deployed in an integrated circuit as an electrostatic discharge protection device. The structure **10** may be used, for example, for either input/output pad protection or power pad protection. Current from an electrostatic discharge event may be received at the terminal **42** and shunted through the lateral silicon-controlled rectifier to the terminal **44**. Alternatively, current from an electrostatic discharge event may be received at the terminal **44** and shunted through the lateral silicon-controlled rectifier to the terminal **42**.

(23) The segments **16** of the discontinuous deep well **14** electrically isolate the well **26** from the semiconductor substrate **12**. The structure **10** with the segmented deep well **14** may exhibit similar performance as an electrostatic discharge protection device to a structure that includes a conventional continuous deep well. The segmented deep well **14**, in comparison with a conventional continuous deep well, may be characterized by a reduced capacitance between the deep well **14** and the well **26** and a reduced capacitance between the deep well **14** and the semiconductor substrate **12**, as well as a lowered harmonics distortion.

(24) With reference to FIGS. **4, 4A** and in accordance with alternative embodiments of the invention, the wells **28, 30, 32** (FIG. **3**) may be joined to define a continuous well **33** that may fully surround the well **26** on all sides, as well as separately surround the well **24** on all sides. The well **33** overlaps with at least a portion of each of the segments **16** of the deep well **14**. Specifically, the well **33** may fully overlap with the peripheral segments **16**, and the well **33** may overlap with opposite end portions of the non-peripheral segments **16**. Processing continues to form the structure **10** as previously described.

(25) With reference to FIG. **5** and in accordance with alternative embodiments of the invention, the segments **16** of the deep well **14** may be further segmented to have a shortened length, and the shortened segments **16** may be distributed in rows with the segments **16** in each separate row aligned along the respective longitudinal axis **18**. In an embodiment, the rows of segments **16** may be arranged parallel to each other. The segments **16** in each row are longitudinally distributed with an end-to-end arrangement, in addition to a side-by-side arrangement segments **16** in adjacent rows. In an embodiment, the segments **16** in each row may be arranged with a constant pitch. Portions of the oppositely-doped semiconductor substrate **12** are positioned in the spaces **49** between the ends of adjacent segments **16** in each row, as well as in the gaps **20** between adjacent pairs of rows. In the representative embodiment, the segments **16** in adjacent pairs of rows are offset or staggered in longitudinal position. Processing continues to form the structure **10** as previously described.

(26) With reference to FIG. **6** and in accordance with alternative embodiments of the invention, the segmented deep well **14** may be used in conjunction with a structure **50** including a diode, which may be deployed as an electrostatic discharge protection device. The structure **50** includes the doped region **36** that may function as an anode of the diode and the doped region **40** that may

function as a cathode of the diode. The segments **16** of the deep well **14** are positioned between the diode and the portion of the semiconductor substrate **12** beneath the deep well **14**.

(27) With reference to FIG. 7 and in accordance with alternative embodiments of the invention, the segmented deep well **14** may be used in conjunction with a structure **52** including a field-effect transistor, which may be deployed as an electrostatic discharge protection device. The structure **52** may include a doped region **41** that is similar or identical to doped region **40** and that is also located in the well **26**. The doped regions **40**, **41** may function as the source and drain of the field-effect transistor, which further includes a gate structure **54** arranged over a portion of the well **26** between the doped region **40** and the doped region **41**. The gate structure **54** may include a gate electrode and a gate dielectric layer formed on the semiconductor substrate **12** by depositing a layer stack and patterning the layer stack with photolithography and etching processes. The gate electrode of the gate structure **54** may be comprised of a conductor, such as doped polycrystalline silicon (i.e., polysilicon) or a work function metal, and the gate dielectric layer of the gate structure **54** may be comprised of an electrical insulator, such as silicon dioxide or hafnium oxide. The doped region **36** may provide a well contact that is coupled to the well **26**. The segments **16** of the deep well **14** are positioned between the field-effect transistor and the portion of the semiconductor substrate **12** beneath the deep well **14**.

(28) With reference to FIG. 8 and in accordance with alternative embodiments of the invention, the segmented deep well **14** may be used in conjunction with a structure **56** including a lateral bipolar junction transistor, which may be deployed as an electrostatic discharge protection device. The doped region **40** and doped region **41** function as the emitter and collector of the lateral bipolar junction transistor, the well **26** functions as the base of the lateral bipolar junction transistor, and the doped region **36** functions as an extrinsic base of the lateral bipolar junction transistor.

(29) The methods as described above are used in the fabrication of integrated circuit chips. The resulting integrated circuit chips can be distributed by the fabricator in raw wafer form (e.g., as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. The chip may be integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either an intermediate product or an end product. The end product can be any product that includes integrated circuit chips, such as computer products having a central processor or smartphones.

(30) References herein to terms modified by language of approximation, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. The language of approximation may correspond to the precision of an instrument used to measure the value and, unless otherwise dependent on the precision of the instrument, may indicate a range of $\pm 10\%$ of the stated value(s).

(31) References herein to terms such as “vertical”, “horizontal”, etc. are made by way of example, and not by way of limitation, to establish a frame of reference. The term “horizontal” as used herein is defined as a plane parallel to a conventional plane of a semiconductor substrate, regardless of its actual three-dimensional spatial orientation. The terms “vertical” and “normal” refer to a direction perpendicular to the horizontal, as just defined. The term “lateral” refers to a direction within the horizontal plane.

(32) A feature “connected” or “coupled” to or with another feature may be directly connected or coupled to or with the other feature or, instead, one or more intervening features may be present. A feature may be “directly connected” or “directly coupled” to or with another feature if intervening features are absent. A feature may be “indirectly connected” or “indirectly coupled” to or with another feature if at least one intervening feature is present. A feature “on” or “contacting” another feature may be directly on or in direct contact with the other feature or, instead, one or more intervening features may be present. A feature may be “directly on” or in “direct contact” with another feature if intervening features are absent. A feature may be “indirectly on” or in “indirect contact” with another feature if at least one intervening feature is present. Different features may

“overlap” if a feature extends over, and covers a part of, another feature.

(33) The descriptions of the various embodiments of the present invention have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A structure comprising: a semiconductor substrate having a top surface; a first well of a first conductivity type in the semiconductor substrate, the first well including a plurality of segments; a second well of a second conductivity type in the semiconductor substrate, the second conductivity type different from the first conductivity type, the second well positioned in a vertical direction between the segments of the first well and the top surface of the semiconductor substrate, and the second well overlapping with the segments; a third well of the first conductivity type in the semiconductor substrate, the third well positioned in a lateral direction adjacent to the second well; and a fourth well of the first conductivity type in the semiconductor substrate, the fourth well positioned in the lateral direction adjacent to the second well, wherein the second well is positioned in the lateral direction between the third well and the fourth well.
2. The structure of claim 1 wherein the segments comprise a plurality of strips positioned with a spaced arrangement in the lateral direction.
3. The structure of claim 2 wherein each adjacent pair of the segments is separated in the lateral direction by a strip of the semiconductor substrate.
4. The structure of claim 2 wherein the segments are arranged parallel to each other.
5. The structure of claim 1 wherein the segments are longitudinally distributed with an end-to-end arrangement in a plurality of rows.
6. The structure of claim 5 wherein adjacent pairs of the rows are arranged parallel to each other.
7. The structure of claim 5 wherein each row is aligned along a longitudinal axis, and the segments in each row are separated in a longitudinal direction by a portion of the semiconductor substrate.
8. The structure of claim 5 wherein the segments in adjacent pairs of the rows are staggered in longitudinal position.
9. The structure of claim 1 further comprising: a silicon-controlled-rectifier including a first doped region of the first conductivity type in the second well and a second doped region of the second conductivity type in the second well.
10. The structure of claim 1 further comprising: a diode including a first doped region of the first conductivity type in the second well and a second doped region of the second conductivity type in the second well.
11. The structure of claim 1 further comprising: a field-effect transistor including a first doped region of the first conductivity type in the second well and a second doped region of the first conductivity type in the second well.
12. The structure of claim 1 further comprising: a lateral bipolar junction transistor including a first doped region of the first conductivity type in the second well and a second doped region of the first conductivity type in the second well.
13. A method comprising: forming a first well of a first conductivity type in a semiconductor substrate, wherein the first well includes a plurality of segments; forming a second well of a second conductivity type in the semiconductor substrate, wherein the second conductivity type is opposite to the first conductivity type, the second well is positioned in a vertical direction between the segments of the first well and a top surface of the semiconductor substrate, and the second well

overlaps with the segments forming a third well of the first conductivity type in the semiconductor substrate, wherein the third well is positioned in a lateral direction adjacent to the second well; and forming a fourth well of the first conductivity type in the semiconductor substrate, wherein the fourth well is positioned in the lateral direction adjacent to the second well, and the second well is positioned in the lateral direction between the third well and the fourth well.

14. The method of claim 13 further comprising: forming a silicon-controlled-rectifier including a first doped region of the first conductivity type in the second well and a second doped region of the second conductivity type in the second well.

15. The method of claim 13 wherein the segments comprise a plurality of strips positioned with a spaced arrangement in the lateral direction, and each adjacent pair of the segments is separated in the lateral direction by a strip of the semiconductor substrate.
